

42. Between two distinct rational numbers a and b , there exists another rational number which is (P.C.S., 2006)
- (a) $\frac{a}{2}$ (b) $\frac{b}{2}$
 (c) $\frac{ab}{2}$ (d) $\frac{a+b}{2}$
43. If $B > A$, then which expression will have the highest value (given that A and B are positive integers)? (Campus Recruitment, 2007)
- (a) $A - B$ (b) AB
 (c) $A + B$ (d) Can't say
44. If $0 < x < 1$, which of the following is greatest? (Campus Recruitment, 2007)
- (a) x (b) x^2
 (c) $\frac{1}{x}$ (d) $\frac{1}{x^2}$
45. If p is a positive fraction less than 1, then
- (a) $\frac{1}{p}$ is less than 1 (b) $\frac{1}{p}$ is a positive integer
 (c) p^2 is less than p
 (d) $\frac{2}{p} - p$ is a positive number
46. If x is a real number, then $x^2 + x + 1$ is
- (a) less than $\frac{3}{4}$
 (b) zero for at least one value of x
 (c) always negative
 (d) greater than or equal to $\frac{3}{4}$
47. Let n be a natural number such that $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{n}$ is also a natural number. Which of the following statements is not true? (A.A.O. Exam, 2009)
- (a) 2 divides n (b) 3 divides n
 (c) 7 divides n (d) $n > 84$
48. If n is an integer, how many values of n will give an integral value of $\left(\frac{16n^2 + 7n + 6}{n}\right)$?
- (a) 2 (b) 3
 (c) 4 (d) None of these
49. If $p > q$ and $r < 0$, then which is true?
- (a) $pr < qr$ (b) $p - r < q - r$
 (c) $p + r < q + r$ (d) None of these
50. If $X < Z$ and $X < Y$, which of the following is necessarily true?
- I. $Y < Z$ II. $X^2 < YZ$
 III. $ZX < Y + Z$
 (a) Only I (b) Only II
 (c) Only III (d) None of these
51. In the relation $x > y + z$, $x + y > p$ and $z < p$, which of the following is necessarily true? (Campus Recruitment, 2008)
- (a) $y > p$ (b) $x + y > z$
 (c) $y + p > x$ (d) Insufficient data
52. If a and b are positive integers and $\frac{(a-b)}{3.5} = \frac{4}{7}$, then (Campus Recruitment, 2010)
- (a) $b > a$ (b) $b < a$
 (c) $b = a$ (d) $b \geq a$
53. If $13 = \frac{13w}{(1-w)}$, then $(2w)^2 = ?$ (Campus Recruitment, 2009)
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) 1 (d) 2
- Directions (Questions 54–57):** For a 5-digit number, without repetition of digits, the following information is available. (B.B.A., 2006)
- (i) The first digit is more than 5 times the last digit.
 (ii) The two-digit number formed by the last two digits is the product of two prime numbers.
 (iii) The first three digits are all odd.
 (iv) The number does not contain the digits 3 or 0 and the first digit is also the largest.
54. The second digit of the number is
- (a) 5 (b) 7
 (c) 9 (d) Cannot be determined
55. The last digit of the number is
- (a) 0 (b) 1
 (c) 2 (d) 3
56. The largest digit in the number is
- (a) 5 (b) 7
 (c) 8 (d) 9
57. Which of the following is a factor of the given number?
- (a) 2 (b) 3
 (c) 4 (d) 9
58. The least prime number is
- (a) 0 (b) 1
 (c) 2 (d) 3
59. Consider the following statements:
1. If x and y are composite numbers, then $x + y$ is always composite.
 2. There does not exist a natural number which is neither prime nor composite.
 Which of the above statements is/are correct?
- (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2
60. The number of prime numbers between 0 and 50 is
- (a) 14 (b) 15
 (c) 16 (d) 17